

Agent as a Service (AaaS) — Technical Architecture Blueprint

Autonomous Multi-Agent Control Plane, JIT Ephemeral Access Brokering & Agent Triad Framework

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1. RECONSTRUCTED SYSTEM ARCHITECTURE BLUEPRINT

The unified AaaS platform enforces a strict decoupling between the **Central SaaS Control Plane** (agent reasoning, multi-agent arbitration, and JIT token orchestration) and the **Customer GCP Data Plane** (read-only telemetry extraction, isolated execution webhooks, and workload resources).

GCP CENTRAL SAAS CONTROL PLANE

Multi-Tenant UI Dashboard

- Incident Triage View & Live Topology Map
- RCA / Proposed Fix Verification Pane
- HITL Authorization Gate:** Approves Scope & JIT minting

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A2A

ADK Multi-Agent Core (Agent Triad)

- Frontline Watcher:** Perception & Edge Noise Filtering
- Troubleshooting Agent:** Gemini Reasoning & RCA
- Execution Orchestrator:** JIT Minting & Auto-Rollback

2. REASONING & 3. HITL APPROVAL (A2A Protocol)

4. CONTROLLED EXECUTION (OIDC via Workload Identity Fed + JIT SA Token) ↓

▲ 1. INGESTION & DETECTION (Telemetry Streaming) | 5. POST-VERIFICATION WATCHDOG LOOP (Zero-Trust Feedback) ▲

CUSTOMER GCP DATA PLANE (TENANT BOUNDARY)

Telemetry & Observation Layer

- Cloud Logging Sinks (Edge Heuristic Filter)
- Cloud Monitoring Alert Policies & Metric Streams
- Read-Only Least-Privilege SA (Non-Mutating)

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Remediation & Safety Layer

- Executioner Webhook (Cloud Run / Terraform Worker)
- OPA / Rego Parameter Validator & Sanitizer
- Ephemeral JIT Service Account:** 15-min max TTL

Customer Workload Resources: GKE Clusters • Compute Engine VMs • Cloud SQL Databases • Cloud Storage Buckets • Cloud Spanner

2. SECTION 6.1: CURRENT MODEL VS. PROPOSED AAAS MODEL (THE AGENT TRIAD)

Traditional cloud support depends on serialized human triage, causing extensive context loss, placeholder update fatigue, and high MTTR. The **Agent Triad** operational model decomposes workload remediation into three coordinated agents operating under the open **Agent2Agent (A2A)** protocol:

Agent Triad Component	Primary Role & Access Tier	Operational Mechanics & Safety Controls
1. Frontline Watcher Perception Agent	Read-Only SA Non-mutating scope	Continuously polls telemetry streams, filters routine events at the edge via heuristic scorers, and feeds enriched context into Vertex AI Vector Search upon anomaly detection.
		Runs RCA using Gemini Pro on Vertex AI, queries topology baselines in Spanner Graph, resolves operational conflicts

Agent Triad Component	Primary Role & Access Tier	Operational Mechanics & Safety Controls
2. Troubleshooting Agent Reasoning Agent	Memory Fabric & Knowledge Base Querying	(Reliability > FinOps > Advisory), and compiles deterministic dry-run plans.
3. Execution Orchestrator Action & JIT Minting	Ephemeral JIT Token Zero-Standing Privilege	Validates plans through OPA/Rego gates, mints a short-lived JIT service account token via Workload Identity Federation (15-min TTL), applies canary execution, and triggers atomic rollback if regressions occur.

Incident Resolution Flow Comparison

Dimension	Traditional GCP Support Model	Proposed AaaS Model (Agent Triad)
Trigger & Triage	Manual ticket logging via ITSM; category tree navigation; context lost at L1/L2 handoffs.	Autonomous event ingestion via Cloud Pub/Sub; instant anomaly synthesis by Frontline Watcher.
Credentials & Access	Persistent, static IAM service accounts or broad engineer access credentials.	Zero-Standing Privilege: Short-lived, ephemeral JIT token minted per incident via WIF.
Execution Safety	Manual script execution; unverified blast radius; risk of human error during live incidents.	Pre-execution state snapshots, OPA/Rego boundary policies, and isolated canary rollouts.
Verification & Close	Placeholder updates; tickets closed under unresponsive customer protocols without system validation.	Closed-loop post-verification; BRC-ARE Telemetry Watchdog auto-reverts regressions.

3. APPENDIX B: COMPREHENSIVE ARCHITECTURE & MODEL GLOSSARY

A2A (Agent2Agent Protocol)

An open communication protocol enabling structured intent negotiation, distributed context passing, and task delegation across specialized autonomous AI agents.

ADK (Agent Development Kit)

Google Cloud's code-first framework for engineering, evaluating, and chaining multi-agent systems natively on Vertex AI.

BRC-ARE (Blast Radius & Auto-Rollback)

Automated safety engine taking pre-execution state snapshots (Git SHA/ specs) and triggering instant atomic rollbacks if telemetry watchdog limits are breached.

Model Armor

Runtime security layer on Vertex AI that inspects agent prompts and completions against prompt injection, jailbreaks, and sensitive data leakage.

OPA / Rego Engine

Deterministic policy engine validating all IaC patches, configuration mutations, and API payloads against security boundaries before execution.

Agent Triad

The foundational operational triad comprising the Frontline Watcher (Perception), Troubleshooting Agent (Reasoning), and Execution Orchestrator (Action & Token Minting).

JIT (Just-In-Time) Access Broker

Zero-trust security subsystem generating time-bound, cryptographically signed service account tokens via WIF, enforcing Zero-Standing Privileges.

Gemini Models (Flash & Pro)

Google's multimodal foundation models powering high-speed telemetry perception (Flash) and deep root-cause synthesis/tool use (Pro) on Vertex AI.

WIF (Workload Identity Federation)

Identity bridge allowing SaaS Control Plane agents to authenticate into customer tenant projects securely without managing shared keys.

Context & Memory Fabric

Distributed state repository built on Vertex AI Vector Search, Spanner Graph, and BigQuery to retain cross-incident topology and runbook history.

Governance Note: In accordance with Google's Secure AI Framework (SAIF), all mutating actions affecting production systems or regulated data boundaries (PCI-DSS, HIPAA) mandate **Human-in-the-Loop (HITL)** approval via the Central SaaS UI before JIT token issuance.